

PROJECT REPORT

DairyFeed Guardian

AI-Assisted Smart Feed and Silage Quality Screening System for Dairy Farmers

Project domain	Dairy farming, animal nutrition, computer vision, IoT and digital traceability
Primary users	Dairy farmers, dairy cooperatives, feed suppliers and extension workers
Core objective	Provide rapid preliminary visual screening, storage-condition tracking and practical farmer guidance while clearly flagging tests that require validated sensors or laboratory analysis.
Report basis	Prepared from the project brief and the application/report screenshots shared in this conversation. Source code, model files, dataset, and live database were not available for independent technical verification.

Important: This is a project design and implementation report, not a certificate of feed safety, laboratory analysis, veterinary diagnosis, or proof of model accuracy. All performance claims must be supported by testing on the actual trained model and held-out dataset.

Prepared: September 2026

1. Executive Summary

DairyFeed Guardian is proposed as a farmer-facing digital system that combines image-based feed screening, feed-storage monitoring, nutrition guidance, screening history and QR-linked traceability. The interface shown in the supplied screenshots includes Dashboard, Test New Feed, Feed Storage, Nutrition Advisory, Feed Passport, History and Cooperative modules. The concept is intended to make preliminary checks more accessible to dairy farmers and help them identify when a feed sample should be withheld and referred for physical inspection or laboratory testing.

The system must separate what can be inferred from an image from what requires a calibrated sensor or certified laboratory method. A phone camera may support visual classification of visible material or mould-like appearance, but it cannot directly quantify crude protein, fibre, energy, minerals, pH, aflatoxins or mycotoxins. These values must come from validated instruments, laboratory results, or clearly labelled sample/demo records—not from unsupported AI estimates.

The screenshots show the application displaying confidence scores, class probabilities, warnings, storage readings, and advisory content. They also show problems that need to be addressed before a reliable release: demo/seeded profile branding in the real farmer interface, inconsistent or possibly hardcoded dashboard counts, a low-confidence/inconclusive result workflow, missing or blank report content, and unreadable Hindi glyphs in exported PDFs.

This report specifies the intended architecture, workflows, data handling, validation plan, safety controls and acceptance criteria. It does not claim that these fixes have already been implemented or that the model has been independently evaluated.

2. Problem Statement and Objectives

Dairy farmers may encounter variable feed quality, adulteration, moisture-related spoilage, mould growth, poor silage fermentation and limited access to timely laboratory testing. Delayed or incorrect decisions can affect animal health, milk production and farm economics. A portable, low-cost digital workflow can help record observations, screen visible risks and route uncertain or high-risk samples to appropriate follow-up.

Project objectives

- Provide a simple mobile-friendly workflow to capture or upload a feed/silage sample image.
- Run the actual trained image model and display its predicted class, confidence, model/version identifier and uncertainty status.
- Prevent low-confidence, out-of-distribution or unsupported samples from being presented as confirmed safe.
- Record storage measurements with unit, timestamp, location, source and sensor/manual provenance.
- Generate clear, multilingual, conservative farmer guidance and referrals for laboratory or veterinary support.
- Create screening history and QR-based sample records using persistent data rather than fabricated dashboard totals.
- Produce readable, printable reports with embedded fonts and verified sample data.

3. System Scope and Functional Modules

MODULE	REQUIRED BEHAVIOUR
Dashboard	Summarize persisted screening records, risk categories and recent activity. Every count must be calculated from the database and labelled by source and time range.
Test New Feed	Capture/upload image; validate file type and image quality; run actual inference; show top class probabilities, threshold, uncertainty and recommended next action.
Feed Storage	Register silo/bunker/store units; log sensor or manual temperature, humidity, gas and inspection readings; show units, timestamp, source and configured threshold.
Nutrition Advisory	Collect animal profile and measured feed inputs; show ration guidance only within documented assumptions and validated calculation rules. Mark missing lab measurements explicitly.
Feed Passport	Create a unique sample/batch ID and QR code linking to the stored record, source, date, test status and traceability events.
History	Display saved screenings and storage inspections with filters, detail view, export and correction/audit history.
Cooperative	Support authorized cooperative-level batch records, aggregation and traceability with role-based access and consent.

4. AI Model and Prediction Integrity

The supplied UI screenshot shows a MobileNetV2/PyTorch model label and example class probabilities for categories such as foreign non-feed debris, fungal/mould colonization and normal/fresh feed grain. These are screenshot-level observations only; the model file, class mapping, training dataset, preprocessing pipeline and evaluation results were not supplied for inspection.

Required inference pipeline

- Load the trained model artifact and its exact class-index mapping from a versioned configuration. Fail clearly if the artifact or mapping is missing.
- Apply the same image resizing, normalization, channel order and preprocessing used during training.
- Return class probabilities from the model, predicted class, model version, inference timestamp and input image reference.
- Apply a validation-backed confidence threshold and an out-of-distribution/quality check. If the image is blurred, poorly lit, unsupported, or uncertain, return an inconclusive result and request retake/manual inspection.
- Do not label a sample safe solely because the normal class has the largest probability. A confidence score is not a calibrated probability of safety unless calibration has been evaluated.
- Log model errors and distinguish a genuine model result from a mock/demo response. Never silently fall back to mock data in Real AI mode.

Model validation requirements

Evaluate using a held-out test set that was not used for training or tuning. Report class-wise precision, recall, F1-score, confusion matrix, class counts, dataset provenance, split method and known limitations. Include performance across feed types, lighting, camera devices, crop varieties and sample preparation conditions where represented. Do not advertise accuracy until the actual evaluation is completed and documented.

5. Measurement Capabilities and Limitations

Parameter / risk	Possible source	Required user-facing label
Visible foreign objects / visual mould-like appearance	Image model, within validated classes	AI visual screening; not definitive contamination diagnosis
Moisture	Calibrated moisture sensor or lab method	Measured only when instrument reading is supplied; otherwise Not measured
Temperature / relative humidity	Connected, calibrated sensor or manual entry	Show source, unit, timestamp and calibration status
pH / fermentation acids	Appropriate probe or laboratory analysis	Not available from ordinary RGB image
Crude protein, fibre, energy, minerals	Validated NIR device or laboratory assay	Do not infer as measured from a phone photo
Aflatoxin / mycotoxins	Validated rapid test kit or accredited laboratory method	Suspected risk requires testing; image cannot confirm toxin presence

6. Storage Monitoring and Advisory Logic

The storage screen shown in the supplied application includes silo/bunker cards with temperature, humidity and CO₂-like readings, and source labels such as Manual or Hardware. The application must not imply hardware telemetry when a value was typed or generated for demonstration.

Each reading should store: facility/unit ID, feed type, value, unit, timestamp with timezone, source type (sensor/manual/demo), sensor identifier where applicable, calibration/maintenance status, user who recorded it, and any notes. Thresholds must be configurable for the sensor location and feed type, supported by documented operating guidance, and shown with the basis or source. Avoid universal claims such as one temperature threshold guaranteeing safe silage.

Advisory workflow

- Low confidence, blurry image or unfamiliar sample: request a new image under even daylight; if still inconclusive, recommend physical inspection.
- Visible mould, unusual heating, putrid odour, suspected adulteration or other concerning signs: advise not to feed the suspect batch until it has been assessed by a qualified person.
- Possible toxin contamination: recommend representative sampling and a validated test or qualified laboratory. Do not report a toxin concentration from image classification.
- Ration and animal-health advice: use animal weight, milk yield, lactation stage, available forage and verified feed analysis. State assumptions and refer individualized or clinical decisions to a veterinarian or dairy nutrition professional.

7. Data Architecture, Privacy and Traceability

Recommended core records

Entity	Key fields
Farmer	farmer_id, display name, preferred language, village/location (with consent), role
Sample / screening	screening_id, farmer_id, sample type, image reference, created_at, model version, prediction, confidence, status, provenance
Model evaluation	model version, dataset version, class map, metrics, evaluation date, limitations
Storage unit / reading	unit_id, feed type, reading, unit, timestamp, source, sensor ID, calibration status
Lab result	sample_id, laboratory/test provider, analyte, result, unit, method, report reference, test date
Audit event	record ID, actor, action, timestamp, old/new value where appropriate

Use authentication and role-based access for farmer, cooperative and administrator functions. Collect only necessary personal data, obtain consent for location and sharing, protect uploaded images and reports, and provide a way to correct or delete records according to project policy. QR codes should expose only the intended shareable batch information—not private farmer data by default.

8. PDF Report and Hindi Localization Requirements

The shared exported report screenshot shows black square glyphs in Hindi text and blank or incomplete-looking fields. The likely category of issue is font coverage/shaping or a PDF generation/rendering problem; the source code and original PDF were not inspected in this report, so the exact root cause remains unverified.

Implementation requirements

- Use a font with Devanagari coverage (for example, Noto Sans Devanagari) and package the font file with the application.
- Use a PDF renderer that supports Indic shaping and conjunct formation; font embedding alone may not be enough.
- Ensure all report fields are populated from the selected saved screening record. Show Not available / Not measured for absent values, never fabricated data.
- Use A4 page size, consistent margins, readable type, correct line wrapping, repeated table headers, and safe page breaks.
- Render every page to an image during testing; inspect Hindi and English text, tables, footers, symbols and page numbers.
- Verify extracted text and open the exported file in more than one PDF viewer. Keep generated reports linked to their screening ID and generation timestamp.

9. Known Issues Visible in Shared Screenshots

Observed in screenshot	Required correction / verification
“Seeded Demo Profile” appears in the dashboard hero	Remove seeded/demo branding from real farmer experience. Demo mode, if retained for development, must be separate, optional and unmistakably labelled.
Dashboard shows counts and percentages (e.g. screenings, safe, attention, high-risk)	Verify all figures derive from persistent records and defined status rules. Reconcile totals and label date range and source. Do not hardcode production counts.
Real AI / Demo Mode controls and mixed source counts are visible	Real AI mode must call the actual model. If model unavailable, show an explicit error and retry/manual path; never quietly substitute mock predictions.
An example result is inconclusive and recommends manual inspection	Keep this safe fallback. Show calibrated threshold, actual model confidence and clear reason; avoid overstating the confidence as safety probability.
Nutrition page lists missing certified laboratory measurements	Keep missing lab values visibly marked as not measured. Do not manufacture crude protein, NDF, TDN, toxin or fermentation-acid values.
Storage cards mix Manual and Hardware source labels	Ensure each record's source is accurate. Demo/mock readings must never be presented as hardware telemetry.
PDF contains black squares / unreadable Hindi and some blank fields	Fix Devanagari font shaping and data binding, then inspect the entire rendered PDF before release.

10. Implementation and Testing Plan

Phase	Work and exit criteria
1. Audit	Inspect repository, trained model artifact, class labels, dataset splits, database, PDF generator, fonts and environment configuration. Document what is actually connected.
2. Data integrity	Connect dashboard/history/passport/report to persistent records; remove production seed data; add source/provenance fields and explicit error states.
3. Inference reliability	Verify preprocessing and class mapping; add image-quality and uncertainty handling; evaluate on held-out data and publish metrics.
4. Storage & advisory	Connect real sensor integrations where available; clearly mark manual inputs; document threshold basis; keep unsupported analytes as not measured.
5. Localization & PDF	Bundle Unicode fonts; enable Indic shaping; fix layout and field bindings; render and inspect all pages in Hindi and English.
6. Release validation	Run functional, model, security, offline, PDF and usability tests; record test evidence and known limitations before demonstration or deployment.

11. Acceptance Checklist

- ☐ Real AI mode runs the versioned trained model and displays its actual output, or gives an explicit error.
- ☐ Prediction class mapping matches training; confidence and threshold behaviour are tested and documented.

- ☐ Dashboard totals reconcile with persisted records and exclude mock records from real statistics.
- ☐ Every storage value includes units, timestamp and correct manual/sensor/demo source.
- ☐ Unsupported laboratory parameters are marked not measured unless verified test results are attached.
- ☐ Low-confidence and suspected contamination workflows avoid false safe assurances and provide practical next steps.
- ☐ Hindi and English render correctly throughout every PDF page, with no black boxes, clipped text or blank data fields.
- ☐ QR, history, exports and access controls behave correctly with real saved records.
- ☐ Offline behaviour and synchronization conflicts are tested and clearly indicated to users.

12. Conclusion

DairyFeed Guardian has a useful proposed workflow for bringing preliminary feed screening, storage logging, farmer guidance and traceability into one interface. Its readiness depends on connecting the actual trained model and persistent data, validating the model on independent data, clearly separating image screening from laboratory measurement, correcting PDF Hindi rendering, and eliminating demo data from the real farmer experience. Until these checks are complete, the system should be presented as a prototype and its outputs as screening guidance—not as a certified feed-quality or animal-health determination.

Safety and scope disclaimer

This report describes a software project and its validation requirements. It is not a veterinary diagnosis, certified laboratory report, guarantee of feed safety, or proof of model performance. Follow qualified veterinary, dairy nutrition and laboratory guidance for animal feeding and suspected contamination.